== MASTER TUTOR SCRIPT: MHF4U UNIT 2 - DIVISION & THEOREMS Student: Rigden

HOW TO USE THIS SCRIPT: Keep this open on a second monitor or printed out while sharing the presentation with Rigden. It contains the exact flow, expected answers, and troubleshooting tips for all 31 slides.

PART A: UNIT 1 QUICK DIAGNOSTIC

SLIDE 1: Title [SCRIPT]: "Welcome back, Rigden! Today is an exciting day. We are officially moving into Unit 2. But before we learn the new tricks, I want to make sure your Unit 1 foundation is rock solid."

SLIDE 2: Roadmap [SCRIPT]: "Here is our game plan for the next hour. We'll do a rapid-fire check of Unit 1, learn how to divide polynomials manually, learn a massive shortcut called Synthetic Division, and finally, learn how to graph massive functions from scratch."

SLIDE 3: Warm-Up Quiz (End Behaviour) [SCRIPT]: "Let's warm up those math muscles. Look at equation A. Without calculating anything, tell me which quadrants the graph starts and ends in, and why." [EXPECTED ANSWER]: "Degree is 5 (odd), coefficient is negative. It trends down, so Q2 to Q4." [SCRIPT]: "Perfect. Now equation B?" [EXPECTED ANSWER]: "Even degree, positive coefficient -> Q2 to Q1."

SLIDE 4: Factored Form Review [SCRIPT]: "Last week, we loved factored form because it gives us the x-intercepts for free. In this equation, what happens at x=3 and x=-1?" [EXPECTED ANSWER]: "At x=3 it bounces (degree 2). At x=-1 it inflects/flattens and crosses (degree 3)."

SLIDE 5: Graph Interpretation Challenge [SCRIPT]: "Let's work backwards. Look at this graph. Build the skeleton equation for me." [EXPECTED ANSWER]: "y = a(x+2)(x-1)^2" [EXAM TIP (Tell Rigden)]: "Ontario exams love testing this backwards thinking. Always check your end behaviour against your final equation to make sure your 'a' value makes sense!"

SLIDE 6: The Transition (The Monster) [SCRIPT]: "You just proved you can graph anything in factored form. But what if a test gives you this? You can't see the x-intercepts, and you can't factor it using Grade 10 quadratic rules. How do we break this monster apart? We have to divide it. Let's enter Unit 2."

PART B: POLYNOMIAL LONG DIVISION

SLIDE 7: The "Old School" Connection [SCRIPT]: "Before we look at algebra, let's go back to elementary school. How do you solve 125 divided by 5? You use a loop: Divide, Multiply, Subtract, Bring Down. Does McDonald's Sell Burgers?"

SLIDE 8: Setup & Vocabulary [SCRIPT]: "Polynomial division is the exact same loop. The big polynomial is the dividend inside the house. The small binomial is the divisor outside."

SLIDE 9: The D-M-S-B Loop [SCRIPT]: "Let's run the loop together. Divide: How many times does x go into x^3? It goes in x^2 times. Write it up top. Multiply: Distribute x^2 to (x-3) to get x^3 - 3x^2. Subtract: Change the signs! This is where 90% of students mess up. Change them, then add. Bring down: Bring down the -5x." [ASK]: "Rigden, what is our next 'Divide' step?"

SLIDE 10: Division Statement [SCRIPT]: "Once we finish dividing, we write a division statement. It's just a fancy way of saying 13 = 5 x 2 + 3. We just put the polynomial pieces into those slots."

SLIDE 11: Your Turn! (Long Division) [SCRIPT]: "Your turn. Grab your paper and run the D-M-S-B loop. Talk me through your first step." [TUTOR TIP]: Wait for Rigden to solve. Guide him gently if he forgets to change signs during subtraction.

SLIDE 12: The Hidden Trap (Placeholders) [SCRIPT]: "Exam trap alert! If you divide x^3 - 8, you are missing the x^2 and x terms. If you don't put 0x^2 and 0x as placeholders, your columns won't line up and your math will crash. Always check for missing degrees!"

PART C: SYNTHETIC DIVISION

SLIDE 13: Synthetic Division (The Cheat Code) [SCRIPT]: "Long division takes too much time. Let me show you a cheat code: Synthetic Division. It completely removes the x's and only looks at the numbers."

SLIDE 14: Setup & The k-Value [SCRIPT]: "Rule #1: Only use this when dividing by a linear factor like (x-k). Rule #2: Put the root outside, not the factor. If the factor is (x-2), what number goes outside?" [EXPECTED ANSWER]: "2"

SLIDE 15: Drop, Multiply, Add [SCRIPT]: "The new loop is just three steps: Drop, Multiply, Add. Drop the 1. Multiply 1 x 2 = 2. Add it to the next column. Look how incredibly fast this is compared to long division!"

SLIDE 16: Reading the Answer [SCRIPT]: "The numbers at the bottom are your answer. The last number is always the remainder. The numbers before it are your quotient, but they are exactly one degree lower than what you started with. Since we started with x^3, our answer starts with 1x^2."

SLIDE 17: Your Turn! (Synthetic Division) [SCRIPT]: "Your turn to use the cheat code. What number are you putting outside the bracket?" [EXPECTED ANSWER]: "-1" (Praise him and let him run the drop-multiply-add loop).

SLIDE 18: The (ax-b) Trap [EXAM TIP]: "Another massive Ontario exam trick. If the divisor has a number in front of the x, like 2x-1, your root is 1/2. You can use synthetic division with 1/2, BUT you must divide your final quotient numbers by 2 at the end! Remember this for your tests."

PART D: REMAINDER & FACTOR THEOREMS

SLIDE 19: The Remainder Theorem [SCRIPT]: "Sometimes a question will ask 'What is the remainder?' and nothing else. Running a full division is a waste of time. The Remainder Theorem says: When P(x) is divided by (x-k), the remainder is simply P(k)."

SLIDE 20: Visual Proof [SCRIPT]: "Look at this. We did the division on the left and got a remainder of 1. On the right, we just plugged the number 2 into the equation, and got... 1! Which method is faster?"

SLIDE 21: The Factor Theorem [SCRIPT]: "This is the most important theorem in Grade 12 Advanced Functions. Think about normal numbers. If 10 divided by 5 has a remainder of 0, 5 is a perfect factor. In algebra, if you plug in a number and get exactly 0, you just found a perfect factor!"

SLIDE 22: Finding the First Root [SCRIPT]: "How do we find a factor if they don't give us one? We guess and check using factors of the last number (the constant). Let's test P(1) and P(-1). Rigden, calculate P(-1) for me." [EXPECTED ANSWER]: "(-1)^3 - 7(-1) - 6 = -1 + 7 - 6 = 0." [SCRIPT]: "Boom! Because it equals 0, what is the factor?" [EXPECTED ANSWER]: "(x+1)"

SLIDE 23: The Ultimate Boss Fight [SCRIPT]: "This brings us to the final boss. I want you to sketch this polynomial. It's in standard form. We have to break it apart completely."

SLIDE 24: Boss Step 1 [SCRIPT]: "We already did Step 1. We used the Factor Theorem to prove (x+1) is our first piece of the puzzle."

SLIDE 25: Boss Step 2 [SCRIPT]: "Step 2: Use synthetic division to rip that (x+1) out of the main polynomial. Rigden, run the synthetic division for me. Remember your placeholder for x^2!" [EXPECTED ANSWER]: Gets x^2 - x - 6 as the quotient.

SLIDE 26: Boss Step 3 [SCRIPT]: "Incredible. We took a cubic and turned it into a linear factor times a quadratic. Now, use your Grade 10 skills to factor x^2 - x - 6."

SLIDE 27: Boss Step 4 [SCRIPT]: "Look at what you just did. You took a totally un-graphable standard form equation, used theorems to break it down, and sketched it perfectly. This is high-level math."

PART E: CONSOLIDATION & ASSESSMENT

SLIDE 28: Ontario Exam Tricks Summary [SCRIPT]: "Before we wrap up, take a mental picture of this slide. These are the top 3 places students lose marks on MHF4U tests. Don't fall for them."

SLIDE 29: Summary Map [SCRIPT]: "Here is your master toolkit. The Factor Theorem helps you guess the first root. Synthetic division breaks the polynomial down one degree. Then you just factor what's left and sketch!"

SLIDE 30: Exit Ticket [SCRIPT]: "Rapid fire time! Hit me with the answers." [EXPECTED ANSWERS]:

1. Remainder is 7.
2. (x+2) is a factor.
3. -5 goes outside.

SLIDE 31: Homework & Next Steps [SCRIPT]: "You absolutely crushed this session. For homework, I want you working through the Jensen Math worksheets. Practice that synthetic division loop until you can do it in your sleep. Great job today, Rigden!"